DE–730

Sub. Code
13

DISTANCE EDUCATION

M.Sc. (SE) (5 Years Integrated) DEGREE EXAMINATION, MAY 2017.

DIGITAL COMPUTER FUNDAMENTALS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Solve the following as directed :

   (a) Binary to octal and hexadecimal

      (i) 1111111000.0110

      (ii) 11101.1111

      (iii) 010101.0001

   (b) Octal to binary and hexadecimal

      (i) 777.05

      (ii) 601.03

      (iii) 111.76

   (c) Hexadecimal to binary

      (i) CAD.F

      (ii) A9.13

      (iii) ABC.

2. Simplify the following expressions :

   (a) $xy + \overline{x}z + yz = xy + \overline{x}z$

   (b) $\overline{x}y + x\overline{y} + xy + \overline{x}y = 1$

   (c) $(x + y) \cdot (x + z) = x + yz$

   (d) $(x + y) \cdot (x + \overline{y}) = x$.

3. Explain Demultiplexer with block diagrams and gates.

4. Discuss sequential circuits in elementary qualitative treatment.

5. Describe Register and its types.

6. How is clock used and where it is used? Explain.

7. Explain computer basics and its generations.

8. Write short notes on input-output units.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Describe complements and uses with an example.

10. Explain the terms :

    (a) Sum of products

    (b) Product of sums.

11. Give a brief note on Mckluskey method with an example.

12. Discuss Half Adder, Full Adder and Parallel Binary Adder.

13. How to add and subtract parallel arithmetic elements? Explain.

14. Describe gated flip flops and master slave flip flops.

15. Describe generations and classifications of computers in detail.

DE–731

Sub. Code
14

DISTANCE EDUCATION

M.Sc. (SE) (5 Years Integrated) DEGREE EXAMINATION, MAY 2017.

C AND DATA STRUCTURES

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain the various operators supported by C.

2. Write a C program to find the factorial of a given number using recursive function.

3. Explain about defining a structure with a suitable example.

4. How do you create a data file? Explain with example.

5. What is a binary tree? Explain briefly about binary tree representation.

6. Explain any four string functions.

7. Explain about call by value and call by reference.

8. Write briefly about the applications of trees.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the different forms of if statement with suitable example.

10. (a) Discuss the data types in C.

    (b) Explain the for loop with example.

11. Explain the different forms of macro substitution directives with example.

12. Explain in detail about dynamic memory allocation.

13. Discuss in detail about stack and its operations.

14. Explain in detail about doubly linked list.

15. Write a C program to arrange the given numbers in descending order.

DE–732

Sub. Code
15

DISTANCE EDUCATION

M.Sc. (SE) (5 Years Integrated) DEGREE EXAMINATION, MAY 2017.

DISCRETE MATHEMATICS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Write the following statement in symbolic form

   (a) If tigers have wings then the earth travels round the sun.

   (b) Kannan can write the examinations if and only if he pays the fees.

2. Construct the truth table for $\neg(P \vee Q) \wedge (P \vee R)$.

3. In the set of integers we define aRb iff $a+b$ is even Is R is an equivalence relation?

4. If $f : x \to y$ and $g : y \to z$ and if gof is one-to-one then f is one-to-one. Is g onto if gof is on-to?

5. Show that the set of all rationals Q except 1 with the operation * defined by $a * b = a + b - ab, \forall a, b \in Q$ is an abelian group?

6. Let $(S,*)$ be a semigroup. Let (T, Δ) be an algebraic structure. If there exists an on – to mapping $f : s \to T$ such that for any $a, b \in s$, $f(a * b) = f(a) \Delta f(b)$ then show that (T, Δ) is a semi group?

7. Prove that the monoid homomorphism preserves the property of invertibility?

8. Define regular graph. Prove that the sum of the degrees of the points of a graph G is twice the number of edges in G?

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. (a) Show that

$(\neg P \wedge (\neg Q \wedge R)) \vee (Q \wedge R) \vee (P \wedge R) \Leftrightarrow R$ (8)

(b) Determine whether $(\neg Q \wedge (P \rightarrow Q)) \rightarrow \neg P$ is a tautology. (7)

10. (a) Given $S = (1,2,3,4)$ and a relation R on S defined by $R = \{(1,2)(4,3)(2,2)(2,1)(3,1)\}$ show that R is not transitive? (8)

(b) Let R denote a relation on the set of ordered pairs of positive integers such that $(x,y)R(u,v)$ iff $xv = yu$ show that R is an equivalence relation? (7)

11. (a) Prove that composition of functions is associative. (7)

(b) Let $f : A \rightarrow B$ be a bijection. Hence show that $f^{-1} : B \rightarrow A$ is also a bijection and $(f^{-1})^{-1} = f$? (8)

12. Prove that any two right co-sets of H in G are either disjoint or identical.

13. (a) Prove that the intersection of any two normal subgroups of a group is normal subgroup. (8)

(b) prove that every subgroup of an abelian group is normal. (7)

14. (a) Define

(i) Finite graph

(ii) Multi graph

(iii) Pseudo graph

(iv) Simple graph. (5)

(b) Draw all the sample graphs of one, two and three vertices. (5)

(c) In any graph G prove that the number of vertices of odd degree is even. (5)

15. (a) Prove that a graph is a tree iff it is minimally connected. (8)

(b) Prove that a graph with 'n' vertices $(n-1)$ edges and no circuit is connected? (7)

DE–9558

Sub. Code
16

DISTANCE EDUCATION

M.Sc. (SOFTWARE ENGINEERING) (5 Years Integrated)
DEGREE EXAMINATION, MAY 2017.

C PROGRAMMING LAB

Time : Three hours Maximum : 100 marks

One question to be selected by lot system.

Each sub-division carries 50 marks.

1. (a) Using a 'C' Program, calculate the sales commission for the data given below :

Sales value (Rs.)	Commission %
Less than 1000	No commission
Above 1000 but below 2000	5% of sales
Above 2000 but below 5000	8% of sales
Above 5000	10% of sales

(b) Write a C program to find the sum of digits of a given number.

---------- Cut here ----------

2. (a) Write a C program to print Pascal's triangle.

```
    1
   1 1
  1 2 1
 1 3 3 1
1 4 6 4 1
```

(b) Write a C program to find the roots of a quadratic equation.

3. (a) Write a C program to find simple and compound interest given the values of P, N and R.

   (b) Write a C program to sort 'n' numbers in ascending order.

---------- Cut here ----------

4. (a) Write a C program to add two matrices.

   (b) Write a C program to find the factorial of a given number.

---------- Cut here ----------

5. (a) Write a C program to check the given number is Palindrome or not.

   (b) Write a C program to multiply two matrices.

---------- Cut here ----------

6. (a) Write a C program to convert upper case string into lower case string and vice versa.

   (b) Write a C program to count the number of characters and words in a given string.

---------- Cut here ----------

7. (a) Write a C program to convert decimal to binary.

   (b) Write a C program to arrange the given names in alphabetical order.

---------- Cut here ----------

8. (a) Write a C program to compare two strings using without string function.

   (b) Write a C program to check the given year is leap or not.

DE–9559

Sub. Code
17

DISTANCE EDUCATION

M.Sc. (SE) (5 Year's Integrated) DEGREE EXAMINATION, MAY 2017.

DATA STRUCTURES LAB

Time : Three hours Maximum : 100 marks

ONE question should be given to each candidate by lot system.

Each sub-division carries 50 marks.

1. (a) Write a C program to copy the contents of one file to another file.

   (b) Write a C program for liked list implementation of Queue operations.

---------- Cut here ----------

2. (a) Write a C program to implement Push and Pop operations on Stack.

   (b) Write a C program to sort given 10 numbers in ascending order and display the values before and after sorting.

---------- Cut here ----------

3. (a) Write a C program to evaluate the given Mathematical expression using Stack.

   (b) Write a C program to sort the given set of elements using Selection sort.

4. (a) Write a C program to implement insert and delete operations on linked list structure.

   (b) Write a C program to sort the given set of elements using Insertion sort.

---------- Cut here ----------

5. (a) Write a C program to implement insert and delete operations on Queue using Array.

   (b) Write a C program to find an element using Binary search method.

---------- Cut here ----------

6. (a) Write a C program to implement Push and Pop operations on Stack.

   (b) Write a C program to implement insert and delete operations on linked list structure.

---------- Cut here ----------

7. (a) Write a C program to copy the contents of one file to another file.

   (b) Write a C program to sort the given set of elements using insertion sort.

---------- Cut here ----------

8. (a) Write a C program to evaluate the given Mathematical expression using Stack.

   (b) Write a C program for linked list implementation of Queue operations.

DE–733

Sub. Code
23

DISTANCE EDUCATION

M.Sc. (SE) (5 years Integrated) DEGREE EXAMINATION, MAY 2017.

OBJECT ORIENTED PROGRAMMING AND C++

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following.

1. What is an object and a class?

2. Briefly explain the various characteristics of OOPs.

3. What are friend functions? What is the advantage of friend function?

4. What is constructor? Write a program for copy constructor.

5. Explain any two characteristics of pointers.

6. Describe data types in C++.

7. Which operators cannot be overloaded? Write steps to overload + operator so that it can add two complex numbers. Explain with example how can a function template be created.

8. What is a virtual function? Write rules for virtual function. Explain with an example.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR of the following.

9. Difference between object oriented programming and procedure oriented programming.

10. Explain the uses of private and public keywords. How are they different from each other?

11. Explain the characteristics of arrays with example.

12. What are the differences between pointers to constants and constant pointers? Illustrate.

13. Define operator overloading? Explain how to overload unary operator and binary Operator.

14. What does inheritance mean in C++? What are different forms of inheritance? Give an example for each.

15. What is exception? Explain how to handle an exception with an example.

DE–734

Sub. Code
24

DISTANCE EDUCATION

M.Sc. (SE) (5 Years Integrated) DEGREE EXAMINATION, MAY 2017

OPERATING SYSTEMS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain the concept of Processes, Address spaces and Files.

2. Describe the functioning of Layered systems, Client-Server model and Virtual machines in OS structure.

3. Explain the concept of Race conditions, Critical regions, Mutual exclusion and Semaphores in IPC.

4. Discuss on shortest Job First and Policy driven process scheduling algorithms.

5. Describe the working principles of I/O hardware.

6. Explain the concept of Resources and Deadlock modeling.

7. Discuss on Multiprogramming without swapping or paging, and with fixed partitions.

8. Describe the Disk Space management techniques.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. What is an Operating System? Discuss on the history of Operating Systems.

10. Explain the Round-robin, Priority and Multiple queues process scheduling algorithms.

11. Describe the features of I/O Software Layers.

12. Discuss on Deadlock Avoidance and Prevention techniques.

13. Explain the concept of Virtual memory in detail.

14. Describe any four Page replacement algorithms.

15. Discuss on the basics of Files in detail.

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DE–735

Sub. Code
25

DISTANCE EDUCATION

M.Sc. (SE) (5 Years Integrated) DEGREE EXAMINATION, MAY 2017.

RELATIONAL DATABASE MANAGEMENT SYSTEM

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

All questions carry equal marks.

1. Discuss the features of database management system over conventional techniques for handling large data.

2. Write short notes on Data control facilities with suitable examples.

3. Give brief notes on relational database properties.

4. Briefly explain :

   (a) Query within query. (4)

   (b) UNION operator. (4)

5. (a) Write SQL query for the following. (4)

   Use the CREATE TABLE command to create all the tables in the 'Book Database'.

   Branch Table : Branch-no, Branch-name, Branch-location, No-of-employees.

   Publisher table : Publisher-code, Publisher-name, Publisher-city, Publisher-state.

   (b) Differentiate between DROP TABLE command and DELETE command. (4)

6. Write short notes on evaluation of expressions in SQL.

7. Give a brief note on database schema.

8. Briefly explain network system tables.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain in detail :

   (a) Schema and subschema. (8)

   (b) Manipulative capabilities of database. (7)

10. Discuss in detail any two issues encountered in the relational model.

11. Describe the oracle database architecture.

12. Write in detail the various DML commands with appropriate examples.

13. Give a brief note on embedded SQL statements.

14. Explain in detail the various types of outer joins with suitable examples.

15. Write short notes on logical comparison.

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DE–9560

Sub. Code
26

DISTANCE EDUCATION

M.Sc. (SE) (5 Years Integrated) DEGREE EXAMINATION, MAY 207.

C++.LAB

Time : Three Hours Maximum : 100 Marks

ONE question should be given to each candidate by lot system.

Each sub-division carries 50 marks

1. (a) Write a menu driven C++ program to add, subtract two given matrices of order $m \times n$ defined in class, using operator overloading.

   (b) Create a class DONOR that contains donor number, donor name, age, address, sex, blood group. Write a C++ program to print the number, name and address of the donors for the following categories.

      (i) Blood donors having 'A' blood group

      (ii) Blood donors in the age group between 20 to 30.

      (iii) Female donors with 'O+' blood group in the age between 20 to 25.

2. (a) Create a class which consist of EMPLOYEE details like enumber, ename, department, basicpay, grade. Write member functions to get and display them. Derive a class PAY from the above class and write a member function to calculate da, hra, pf depending on the grade and display the payslip in a neat format.

(b) Write a C++ program to find the difference and total length of given two various tubes specified in meters and centimeters using operator overloading.

---------- Cut here ----------

3. (a) Write a program to process students marks with the help of classes. The class has private variables, for name, mark1, mark2, mark3. It has two member functions – getdata ( ) to get input, result ( ) to print the results. All subjects mark must be >=50 for pass otherwise fail.

(b) Write a C++ program using polymorphism to calculate the square of any two numbers of type int, float, double and long.

---------- Cut here ----------

4. (a) Create a class employee that contains employee number, employee name and address. Write a menu driven C++ program to get the details of '*n*' number of employees and display all details in employee namewise sorted order.

(b) Write a C++ program to create a class STRING. Write a member function to initialize, get and display strings. Overload the operator '+' to concatenate two strings, "= =" to compare two strings.

5. (a) Create a class called TIME that has integer data elements for hours, minutes, seconds. The constructors should initialize these data elements to specified value, if given, and otherwise to 0. A member function should display it in hh:mm:ss format (eg. 11 : 50 : 45). The final member function should add two objects of type TIME passed as arguments.

   (b) Create a class ARITH which consists of a FLOAT and INTEGER variable. Write member functions ADD ( ), SUB ( ), MUL ( ), DIV ( ), MOD ( ) to perform addition, subtraction, multiplication, division and modulus division respectively. Write member functions to get and display values.

6. (a) Assume that you want to generate a table of multiples of any given number. Write a C++ program that allows the user to enter the number and then generates the table, formatting it into ten columns and twenty lines.

   (b) Write a C++ program to find the largest of three numbers using inline function.

---------- Cut here ----------

7. (a) Write a C++ program to create a class called STUDENT with data members Regno, Name and age using inheritance, create the classes UGSTUDENT and PGSTUDENT having fields semester, fees and stipend. Enter the data for atleast 5 students. Find the average age for all UG and PG students separately.

   (b) Write a C++ program to implement function overloading in order to compute power (m, n) where

      (i) m is double and n is int

      (ii) m and n are int.

8. (a) Create a class 'COMPLEX' to hold a complex number. Write a friend function to add two complex numbers. Write a main function to add two complex objects.

(b) Using dynamic constructors, write a C++ program to concatenate two given strings.

DE–9561

Sub. Code
27

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, MAY 2017.

RELATIONAL DATABASE MANAGEMENT SYSTEM – LAB

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE questions to each candidate from lot system.

1. (a) Create oracle table with the table name as STUDENT and do the following

- INSERT the values in the table
- Display the students name who's Rollno. is 5
- Display the students percentage between 70 and 85
- UPDATE the PERCENTAGE=90 WHERE Rollno=3;
- Delete the table

(b) Create table sales_order_details with the following fields s_order_no product_no, description, qty_ordered, qty_disp, product_rate, profit_percent, sell_price, supplier_name.

- Select each row and compute sell_price*.50 and sell_price*1.50 for each row selected.
- Select product_no, profit_percent, Sell_price where profit_per is not between 10 and 20 both inclusive.

- Select product_no, description, profit_percent, sell_price where profit_percent is not between 20 and 30.
- Select the suppliername and product_no where suppliername has 'r' or 'h' as second character

---------- Cut here ----------

2. (a) Create the Table CUSTOMER with the following fields cust_no, Fname, Lname, birth_date, city, mob_no,dept_name, salary. Salary should not be less than 5000.

Solve the following query.

- List of customer by salary wise.
- Display details of customer whose birth date is in year of '1980'.
- Display details of customer whose mobile no start with '99..' series.
- Display name and depart name of customer whose city staffing from 'M'.
- Display count of customer having salary greater than 3000.
- List name of customer having 'A' as 2nd letter in first name.

(b) Create the following tables: PROPERTY with the fields of pno, description, area etc., OWNER with the fields of onername, address, phone etc.,

An owner can have one or more properties, but a property belongs to exactly one owner.

- Insert two records into owner table.
- Insert 2 property records for each owner.

- Update phone no of "Mr. Ragu" to 9123456789.
- Delete all properties from "Karaikudi" owned by "Mr. Kumar".
- List names of owners having property in Erode.

---------- Cut here ----------

3. (a) Create the following tables:

EMP with the fields like eno, name, dno, salary etc.,

PROJECT with the fields like pno, pname, control-dno, budget etc.,

- List the names of emp whose salary is greater than 5000 and less than 20000.
- List the names of projects, controlled by department no 503
- List the details of the projects with minimum budget.
- List the details of the projects with maximum budget.
- List the names of employees, working on computerscience department.

(b) Create the following tables: PERSON with the fields like pnumber, pname, birthdate, income etc.,

AREA with the fields like aname, area_type etc.,

- List the names of all people living in moshi' area.
- List details of all people whose names start with the alphabet 'a' and contains maximum 7 alphabets in it.

- Give the count of people who are born on '18th July 1982'
- Give the count of people whose income is below 10000.

---------- Cut here ----------

4. (a) Create a Table for a Company with the following fields and inserts the values:

Fieldname	Fieldtype	Fieldsize
Company_name	Character	15
Proprietor	character	15
Address	character	25
Suppliername	character	15
No of employees	Number	4
GPpercent	number	6with2decimal places

Queries:

- Display all the records of the company which are in the ascending order of GP percent
- Display the name of the company whose supplier name is 'AAA".
- Display the details of the company whose GP percentage is greater than 20 and orderby GP percentage.
- Display the detail of the company having the employee ranging from 300 to 1000
- Display the name of the company whose supplier is same as like Tata's

(b) Create a table named Student with the following fields and insert the values:

Field name	field type	field size
Student Name	character	15
Gender	character	6
Roll No.	character	10
Department Name	character	15
Address	character	25
Percentage	number	4 with 2 decimal places

Queries:

- Calculate the average percentage of the students.
- Display the names bf the students whose percentage is greater than 80
- Display the details of the student who got the highest percentage.
- Display the details of the students whose percentage is between 50 and 70.
- Display the details of the students whose percentage is greater than the percentage of Roll No = 12CA01

---------- Cut here ----------

5. (a) Create the table PRODUCT with the following fields and insert the values:

Field name	Field type	Field size
Product no	number	6
Product name	character	15
Unit of measure	character	15
Quantity	number	6 with 2 decimal places
Total amount	number	8 with 2 decimal places

Queries:

- Using update statements calculate the total amount and then select the record.

- Select the records whose unit of measure is "Kg"
- Select the records whose quantity is greater than 10 and less than or equal to 20
- Calculate the entire total amount by using sum operation
- Calculate the number of records whose unit price is greater than 50 with count operation.

(b) Write a PL/SQL program for income tax bill preparation.

DE–736

Sub. Code
31

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, MAY 2017.

COMPUTER NETWORKS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions only.

1. Explain briefly about connection oriented and connectionless services.

2. Differentiate between satellites and fiber.

3. Explain the Stop and Wait protocol.

4. Write a short note on HDLC.

5. Discuss about Congestion Prevention Policies.

6. Write a short note on Multiplexing.

7. Explain Transposition ciphers with an example.

8. Discuss briefly about the DNS Name Space.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions only.

9. Explain the uses of Computer Networks.

10. Describe the different Guided Transmission Media.

11. Discuss in detail about Error Detection and Correcting Codes.

12. Explain the Link State Routing algorithm with an example.

13. Describe about Internet control protocols.

14. Explain the working of Remote procedure call.

15. Explain the DES and RSA algorithms with examples.

DE-737

Sub. Code
32

DISTANCE EDUCATION

M.Sc. (SE) (5 Years Integrated) DEGREE EXAMINATION, MAY 2017.

INTERNET AND JAVA PROGRAMMING

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following questions.

1. Write about Domain Name System and E-Mail.

2. Discuss about Online chatting and Usenet Newsgroup.

3. Discuss in detail about the main features of Java.

4. Write short notes on Graphics.

5. Discuss about Java Servlets.

6. How do you create HTML Frames? Give an example.

7. Briefly explain about I/O streams.

8. Write about Java beans.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain in detail the World Wide Web, Internet Search Engine and Web browsers.

10. Write about classes and objects, packages and Interfaces in detail with an example.

11. Discuss about working with windows using AWT controls.

12. List and explain the features of Web page design.

13. Explain about Java Script Functions and arrays with an example.

14. Discuss about AWT controls with suitable diagram.

15. Illustrate the Remote Method Invocation with coding.

DE–738

Sub. Code
33

DISTANCE EDUCATION

M.Sc. (SE) (5 Years Integrated) DEGREE EXAMINATION, MAY 2017.

SOFTWARE ENGINEERING

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. What are the characteristics of linear sequential model?

2. Give short notes on RAD Model.

3. Discuss about all the resources for software project planning.

4. Explain in detail about empirical estimation model.

5. Describe the basic concepts and principles of project scheduling.

6. Explain all the components and structure in requirement specification.

7. Illustrate the uses of architectural design. Explain it.

8. Compare Black box testing with White box Testing.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Briefly discuss about software metrics.

10. Describe the software process and project metrics.

11. Explain the techniques require in estimating the software cost.

12. Explain about risk projection, mitigation, monitoring and Management.

13. Briefly explain the information flow and prototyping.

14. Explain about Interface design and its guidelines.

15. Explain in detail about any three software testing methods.

DE–739

Sub. Code
34

DISTANCE EDUCATION

M.Sc. (SE) (5 Years Integrated) DEGREE EXAMINATION, MAY 2017.

COMPUTER GRAPHICS AND MULTIMEDIA

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Discuss about the categories of I/O devices.

2. Explain the procedure of DDA line drawing algorithm with example.

3. Discuss the properties of basic transformation with matrix representation.

4. Elaborate Cohen Sutherland Line Clipping algorithm with procedure and example.

5. How to represent 3D objects? Explain it.

6. Discuss in detail about the multimedia software.

7. Brief the issues involved in synchronization of multimedia software.

8. List out all the applications of multimedia.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Brief about any two circle generating algorithms with suitable examples.

10. Explain Bezier Curve and its properties with neat diagram. Also discuss its applications.

11. Describe the categories of spline representation and explain each with its descriptions.

12. Discuss the following :

    (a) Projection

    (b) Hidden Surface.

13. Explain all the color models in 3D graphics.

14. Discuss the components of multimedia.

15. Describe the working principle of Video Conferencing and Virtual Reality Systems.

DE–740

Sub. Code
35

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Year Integrated) DEGREE EXAMINATION, MAY 2017.

UNIX AND SHELL PROGRAMMING

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain the architecture of the Unix operating system.

2. Discuss the following commands with example:

   (a) cat (b) rmdir.

3. Give short notes on command substitution.

4. Explain how to use *grep* command with example.

5. Write a note on C shell features.

6. Discuss about argument validation in shell script.

7. Explain about file handling utilities.

8. Write a note on start-up and shutdown scripts in Interactive C shell.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. (a) Describe how Unix is more secured than other operating systems.

   (b) Write a note on vi editor.

10. (a) Give a detailed note on shell / Environment Customization.

    (b) How to use aliases in Unix shell commands? Explain.

11. Summarize in detail about awk command with suitable examples.

12. (a) Explain about basic script concepts in korn shell programming.

    (b) Write a note on eval command with example.

13. Discuss in detail about File Management in C shell programming.

14. (a) Explain the various networking commands in Unix.

    (b) Discuss about text processing utilities and backup utilities.

15. (a) Explain redirection in detail with example.

    (b) What are predefined variables? Give example.

———————

DE–9562

Sub. Code
36

DISTANCE EDUCATION

M.Sc. (SE) (5 YEARS INTEGRATED) DEGREE EXAMINATION, MAY 2017.

INTERNET AND JAVA – LAB

Time : Three hours Maximum : 100 marks

The Examiner has to choose any ONE question.

Each subdivision carries 50 marks.

1. (a) Write a Java program using applet to display any 2 images when 2 buttons in the border layout are clicked. The image should be displayed at the centre of the window.

   (b) Write a Swing program to create Buttons with

   (i) Tool tip text

   (ii) Image

   (iii) Border

   (iv) Short cut key

---------- Cut here ----------

2. (a) Write a Java program using Applet to display the different colors and fonts with text.

   (b) Write a Swing program to create the tabbed panels.

---------- Cut here ----------

3. (a) Write a VBScript to do the following :

   (i) Check the given password

   (ii) Change the existing password.

   (b) Design and publish a web page for an organisation.

4. (a) Create a web page in the format of front page of a newspaper using Text link Align the text With colors.

(b) Write a Java program to create a color as the background and foreground.

---------- Cut here ----------

5. (a) Write a Menu-Drive java program to do database functions using database connectivity Facility. The functions include display edit, insert and delete a records from a table.

(b) Write a Java script to create a Colour palet and display the text in the color chosen from the palet with proper background color.

---------- Cut here ----------

6. (a) Write a VBScript to do the following :

(i) Display the current date.

(ii) Find the difference between the two dates.

(iii) Find the age of a person by providing date of birth.

For all use the date in dd/mm/yy format.

(b) Develop a single page advertisement for a shop to be opened shortly.

---------- Cut here ----------

7. (a) Write a java program to get and display the student mark list.

(b) Write a Java script to create an order Form to select the articles.

---------- Cut here ----------

8. (a) Write a java program to create Book entry system for library with textbox, button, and radio button.

(b) Using at least 15 HTML tags, create a screen with a string "WEB Design.

DE–9563

Sub. Code
37

DISTANCE EDUCATION

M.Sc. (5 Years Integrated) DEGREE EXAMINATION, MAY 2017.

Software Engineering

UNIX AND SHELL PROGRAMMING — LAB

Time : Three hours Maximum : 100 marks

Examiner has to choose any ONE question.

Each subdivision carries 50 marks.

1. (a) Write and execute a shell program to generate leave status report of an organization.

   (b) Write a Menu Driven Shell program for the following :

   (i) List of files

   (ii) Processes of users

   (iii) Today's date

   (iv) Users of system

   (v) Quit to UNIX.

---------- Cut here ----------

2. (a) Write and execute a shell program to send mail to groups of users by extracting their functioned group files.

   (b) Write a shell program which accepts the name of a file from the standard input and then performing the following tests on it :

   (i) Enter the 5 names in a file

   (ii) Sort the names in existing file

   (iii) List unsorted and sorted file

   (iv) Quit.

3. (a) Write a program for four-function calculator by using yacc command.

(b) Write a Menu Driven shell program to perform the following tasks.

(i) Enter the sentence in a file

(ii) Search a given whole word in an existing file

(iii) Quit.

---------- Cut here ----------

4. (a) Write a shell program to sum up the series

1/1! + 1/2! + 1/3! + ... + 1/10!

(b) A Hospital maintains patient details. Write a shell program to list the following.

(i) Patients in a blood groupwise

(ii) Patients in age between 20 to 30

(iii) Exit.

---------- Cut here ----------

5. (a) Write a Menu Driven shell program to copy, edit, rename, delete a file.

(b) Write a shell program to prepare electric bill for domestic consumers.

For first 100 units	–	Rs. 0.75/unit
For next 100 units	–	Rs. 1.50/unit
Above 200 units	–	Rs. 3.00/unit

Prepare the Electricity bill in the following format :

Customer no _____

Customer name _____

Previous reading _____

Current reading _____

Units consumed _____

Charge _____

6. (a) Write a shell program which periodically monitors the disk for the existence of a file and then executes the program once the file has been located. (Using UNTIL statement).

(b) Write and execute a shell program to do the following :

(i) Convert given string in center of given row in BOLD, normal or reverse video.

(ii) Write given string at given row, column in BOLD, normal and reverse video.

(iii) Search a given string using awk with various options.

---------- Cut here ----------

7. (a) Write a shell program which accepts the name of a file from the standard input and then performs the following tests on it :

(i) File existence

(ii) File readable

(iii) File writable

(iv) Both readable and writable.

(b) Write and execute a shell program to print monthly payslips for employees in an organization. Assume your own data and generate accordingly.

---------- Cut here ----------

8. (a) Write a shell program using 3 arguments to take the pattern as well as input and output file names. If the pattern is found, display "Pattern found" else display "Error message" Also check if right number of arguments are entered.

(b) Write and execute a shell program to generate leave status report of an organization.

DE-741

Sub. Code
41

DISTANCE EDUCATION

M.Sc. (SE) (5 Years Integrated) DEGREE EXAMINATION, MAY 2017.

VISUAL PROGRAMMING

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following.

1. Explain in detail about visual Basic Menus.

2. Describe Object and instances in VB.

3. Explain user defined functions in VB.

4. Explain document view structure.

5. Write short notes on Splitter window.

6. Explain the need for ODBC for database connectivity.

7. Write short notes on DLL compare it with static libraries.

8. Write short notes on VC++ components.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR of the following.

9. Explain any five VB Controls with example.

10. Describe MFC file handling.

11. Explain how do you perform event handling in VC++.

12. Write a procedure to create user defined DLL.

13. Explain in detail about MDI environment.

14. Explain File system controls.

15. Explain the features of VBX Controls and their properties with suitable example.

DE–742

Sub. Code
42

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, MAY 2017.

SOFTWARE PROJECT MANAGEMENT AND METRICS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following.

1. What is product quality? Explain the product quality attributes and the metrics used for software quality measurement.

2. Describe the issues in software project management.

3. What are the principles of modern software management?

4. Discuss in detail the planning and operational artifacts.

5. What is milestone of a project? Explain the typical minor milestones in the life cycle of iteration.

6. What are the three levels of software process and their attributes?

7. Explain the iteration planning process in detail.

8. What are the basic parameters of an earned value system? Give its graphical representation.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain briefly the various software project metrics and software process metrics and their attributes.

10. What are the various structural measures used for measuring internal product features of software? Discuss in detail.

11. Define work breakdown structure. Explain the planning guidelines in work breakdown structure.

12. Explain in detail the activities of software management team with its artifacts and responsibilities.

13. What are the primitive components of a software change order? Discuss in detail how the change management is achieved.

14. Describe the statistical techniques that are used for data collection and data analysis. Illustrate with an example.

15. Explain the priorities for tailoring the process framework and compare the process primitive of small and large project team size.

DE–743

Sub. Code
43

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, MAY 2017.

WEB TECHNOLOGY

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following.

1. Discuss in detail about the usage of cascading style sheet in web pages.

2. Explain classes and interfaces in Java beans API.

3. Define servlet and explain about reading servlet parameters.

4. What is the significance of ServletContextAttributeEvent class in javax.servlet package.

5. Write in detail about JSP Page translation and processing phases.

6. Write in detail about Passing Control and Data Between JSP Pages.

7. Discuss about various interfaces and classes of javax.sql package.

8. Discuss elaborately about accessing a database from a JSP page.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. (a) Write the html syntax for the following :

(i) Lists

(ii) Tables

(iii) Images

(iv) Forms

(v) Frames.

(b) Create a webpage page for your college with the above elements.

10. (a) List down the core features of Java Bean.

(b) List and explain any three interfaces and methods in java.beans package.

(c) State the relationships between EJB & Java Bean.

11. (a) With a neat sketch explain servtlet life cycle

(b) Discuss about HTTP request and HTTP response messages.

12. (a) List and explain any three Interfaces and classes in javax.servlet package.

(b) Discuss about cookies-session tracking.

13. (a) Explain how JSP is better than servlet.

(b) Explain the procedure of creating, Installing and running a JSP page.

14. (a) With a neat sketch explain the anatomy of a JSP page.

    (b) Explain in detail the procedure of Sharing Session and Application Data between JSP pages.

15. (a) Discuss the procedure involved in deploying JAVA beans in a JSP page.

    (b) Discuss in detail about Struts framework.

DE–744

Sub. Code
44

DISTANCE EDUCATION

M.Sc. (S/W Engg.) (5 Yr. Integrated) DEGREE EXAMINATION, MAY 2017

DATA WAREHOUSING AND MINING

(2013 Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following.

1. Write detailed notes on various kinds of data preprocessing techniques.

2. Give detailed description about Association Rules in Data Mining.

3. Discuss on any two summarization techniques.

4. Describe about impact of outliers on clustering.

5. Illustrate nearest neighbor clustering algorithm with an example.

6. Write notes on decision tree algorithm.

7. What is Web Mining? Discuss.

8. Explain any one application of Web Mining.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Write an overview about the following :

   (a) Major issues on Data Mining. (8)

   (b) Data mining functionalities. (7)

10. (a) Discuss the process involved in KDD process. (10)

    (b) How to retrieve information using Data Mining Techniques? (5)

11. (a) Describe any two neural networks algorithms used in Data Mining. (8)

    (b) Write detailed notes on Genetic Algorithms. (7)

12. Illustrate various hierarchical algorithms with relevant examples.

13. How to estimate error in Pattern Matching? Discuss.

14. Give detailed description about web Content Mining.

15. Explain in detail about ontology based Web Mining Research.

DE–745

Sub. Code
45

DISTANCE EDUCATION

M.Sc. (SE) (5 years Integrated) DEGREE EXAMINATION, MAY 2017.

DISTRIBUTED COMPUTING

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following questions.

1. Explain the Interaction and Failure models.

2. With a program explain the communication of UDP Datagram.

3. Explain the design issues in RPC.

4. Explain how a new process is created in the OS layer.

5. What is a DNS? Explain.

6. Describe the X.500 Directory Service with a neat diagram.

7. Explain the two phase committed protocol for nested transaction.

8. Explain the flat and nested distributed transactions.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the principles of network with suitable definition.

10. Explain the java RMI with a case study.

11. Briefly explain the architecture of OS.

12. How will you synchronize a physical clock? Explain in detail about it with Berkeley algorithm.

13. Give in detail about the recovery process during the transaction.

14. Explain in detail about the Internet Protocols.

15. Explain in short about the Cryptographic Algorithms available in security.

DE–9564

Sub. Code
46

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, MAY 2017.

VISUAL BASIC AND VC++ LAB

Time : Three hours Maximum : 100 marks

One question should be given to each candidate by lot system.

Each sub-division carries 50 marks.

1. (a) Write a VB program to create a scientific calculator using control arrays.

   (b) Write a MFC program to find fibonacci series using visual C++.

---------- Cut here ----------

2. (a) Write a VB program to perform the following operations in a record of random access file :

   (i) insert (ii) delete (iii) search.

   (b) Write a MFC program to find the factorial of given number using visual C++.

---------- Cut here ----------

3. (a) Write a VB program to perform matrix multiplication.

   (b) Write a MFC program to insert an active-X control using VC++.

4. (a) Write a VB program to perform the following string operations:
   (i) length
   (ii) upper case
   (iii) lower case
   (iv) replace
   (v) concate
   (vi) substring.

   (b) Write a MFC program to create a window of desired size using VC++.

---------- Cut here ----------

5. (a) Write a VB program to perform free hand drawing.

   (b) Write a MFC program to create list box in a window using VC++.

---------- Cut here ----------

6. (a) Write a VB program to find fibonacci series.

   (b) Write a MFC program to implement matrix multiplication using VC++.

---------- Cut here ----------

7. (a) Using MS flex grid control display the multiplication and addition table of 20 rows and columns.

   (b) Write an visual C++ program to fill background of the client area with a bit map.

---------- Cut here ----------

8. (a) Write a VB program to find whether a given text (string) is palindrome or not.

   (b) Write a visual C++ program to get the status of the shift and toggle keys using MFC.

DE–9565

Sub. Code
47

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, MAY 2017.

WEB TECHNOLOGY — LAB

Time : Three hours Maximum : 100 marks

Examiner should select and give One question to each candidate to Lot system.

1. (a) Create a HTML table with rows and columns and split them using Rowspan and colspan.

   (b) Write an XML document to display your biodata. Write an XSL style sheet and attach that to the XML document. Validate the document using DTD or XSD.

---------- Cut here ----------

2. (a) Write an ASP program to prepare Employee pay bill using Java Script.

   (b) Create a web page in the format of front page of a news paper using Text links. Align the text with colors.

---------- Cut here ----------

3. (a) Write an ASP program to prepare student performance evaluation document using Java script.

   (b) Write a servlet program to display “Hello World”.

4. (a) Create an image animation using Java script.

   (b) Write Servlet program to send Email message.

---------- Cut here ----------

5. (a) Write a JSP program to display student mark list using JDBC.

   (b) Develop an Application for JSP – Servlet communication.

---------- Cut here ----------

6. (a) Create a HTML table with rows and columns and split them using Rowspan and colspan.

   (b) Create a web page in the format of front page of a news paper using Text links. Align the text with colours.

---------- Cut here ----------

7. (a) Write an XML document to display your biodata. Write an XSL style sheet and attach that to the XML document. Validate the document using DTD or XSD.

   (b) Write an ASP program to prepare Employee pay bill using Java Script.

---------- Cut here ----------

8. (a) Write a servlet program to display "Hello World".

   (b) Create an image animation using Java script.

DE–746

Sub. Code
51

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Yr. Integrated) DEGREE EXAMINATION, MAY 2017.

SOFTWARE TESTING AND REUSE

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Discuss how tests can be designed from a functional or a structural point of view.

2. Discuss on path predicates.

3. What do you mean by ugly domains? How programmers and testers treat them?

4. Discuss on static and dynamic anomaly detection.

5. Discuss on the basic concepts of path products.

6. Write a note on state graphs.

7. Explain how decision tables are used as a basis for test case design.

8. Explain the usage of Jmeter and win runner tools for functional/regression testing.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Discuss on path sensitizing.

10. Describe a model for testing.

11. Discuss on transaction flow testing techniques.

12. Explain the strategies adopted in data flow testing.

13. Explain how KV charts are used in logic based testing.

14. Describe in detail about state testing.

15. Write note on

    (a) Synchronization of test case

    (b) Rapid testing.

DE–747

Sub. Code
52

DISTANCE EDUCATION

M.Sc. (Software Engineering) (5 Years Integrated) DEGREE EXAMINATION, MAY 2017.

CRYPTOGRAPHY AND NETWORK SECURITY

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. What is network security? Give examples of security services.

2. What do you mean by passive and active attacks? Discuss with examples.

3. What is encryption? Discuss the types of attacks on encrypted messages.

4. Write a short note on cipher block modes.

5. Suggest situations in which message authentication without confidentiality is preferable.

6. What is public key encryption? Discuss on the requirements for public-key cryptography.

7. What is PGP? Describe the various PGP services.

8. What is a virus? Write short note on the types of viruses.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Discuss on Internet standards and RFCs.

10. (a) What is DES? Discuss on DES encryption algorithm.

    (b) Write a note on TCP session hipacking.

11. Discuss on the following :

    (a) Message authentication using a one-way hash function

    (b) HMAC.

12. With examples discuss any one public-key cryptography algorithm.

13. Write short note on X.509 authentication service.

14. Discuss on the following :

    (a) Transport layer security

    (b) Dual signature.

15. What is a firewall? Explain the different types of firewalls.

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DE–748

Sub. Code
53

DISTANCE EDUCATION

M.Sc. (SE) (5 Years Integrated) DEGREE EXAMINATION, MAY 2017.

SOFTWARE QUALITY ASSURANCE AND STANDARDS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following.

1. Explain GQM model.

2. Write short notes on quality measurement.

3. Explain quality tasks.

4. Describe SQA plan.

5. Explain case tools on software quality control.

6. Write short notes on reliability models.

7. Explain the elements of QMS.

8. Explain the need for quality standards.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain hierarchical models of Boehm and McCall.

10. Discuss implementation and documentation of SQA.

11. Explain basic quality control tools in detail.

12. Briefly explain Rayleigh model framework on QMS.

13. Discuss complexity metrics and model.

14. Write detailed notes on ISO 9000 series of quality management standards.

15. Give detailed account of CMM.

DE–749

Sub. Code
54

DISTANCE EDUCATION

M.Sc. (SE) (5 Years Integrated) DEGREE EXAMINATION, MAY 2017.

MOBILE COMMUNICATIONS

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following questions.

1. Distinguish between CDMA and FDMA.

2. Discuss about digital video broadcasting.

3. Explain about Radio Access Layer.

4. Write about Wireless ATM.

5. Write the assumptions and goals of Mobile IP.

6. What is reverse tunneling?

7. What is timeout freezing while transmission?

8. What is session protocol?

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Give in detail comparison of various access mechanisms.

10. Explain the following telecommunication systems :

    (a) GSM

    (b) TETRA

    (c) UMTS

11. Explain with suitable example about infrastructure networks and Adhoc networks.

12. Discuss about Wireless ATM working group and its services.

13. Explain the following :

    (a) IPV6

    (b) DHCP

14. Traditional TCP and Indirect TCP - Compare.

15. Explain the following protocols :

    (a) Datagram Protocol

    (b) Transaction Protocol.

DE-750

Sub. Code
55

DISTANCE EDUCATION

M.Sc. (SE) (5 Years Integrated) DEGREE EXAMINATION, MAY 2017.

OPEN SOURCE ARCHITECTURE

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following.

1. What are the advantages of open source software? Explain.

2. Describe any four basic shell commands.

3. Write short notes on the OS bookup process.

4. Briefly explain about web servers.

5. What is GNU Compiler? Explain.

6. Explain about C preprocessor directives.

7. What do you mean by object archiver? Explain.

8. What is graphical debuggers? Explain.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain about configuring disk partitions, file systems and install a GNU/Linux distribution.

10. Explain the basics of TCP/IP networking and routing.

11. Write short notes on following :

    (a) Setting up email servers

    (b) Setting up proxy services.

12. Explain how do you configure X Window system configuration and utilities.

13. Explain about source code versioning and management tools.

14. (a) Explain about statically linked binaries and libraries.

    (b) Explain about dynamically linked libraries.

15. Explain about programming GUI applications with localization support.

DE–9566

Sub. Code
56

DISTANCE EDUCATION

M.Sc. (SE) DEGREE EXAMINATION, MAY 2017.

Software Engineering (5 Years Integrated)

OPEN SOURCE LAB

Time : Three hours Maximum : 100 marks

Examiner should select and give one question to each candidate by lot system.

1. (a) Write a server side PHP program that displays marks, total, grade of a student in tabular format by accepting user inputs for name, number and marks from a HTML form.

   (b) Write a shell program to change the extension of a given file.

---------- Cut here ----------

2. (a) Write a PHP program to create the college application form of the college.

   (b) Write a PHP program to insert the data to my sql table.

---------- Cut here ----------

3. (a) Write a PHP program to upload a file to the server.

   (b) Write a PHP program to access the data from my sql table and Display the data.

4. (a) Write a PHP program to insert, update and delete library information from the database.

   (b) To create a new PHP program for creating Login form using controls.

---------- Cut here ----------

5. (a) Write a PHP program to create a director, and to read contents from the directory.

   (b) Write a shell program to find the details of a user session.

---------- Cut here ----------

6. (a) To develop a PHP program for creating simple web site.

   (b) To create a PHP script for accessing the system date and time and image to display in the browser.

---------- Cut here ----------

7. (a) Create a PHP program to count the page view using session functions.

   (b) Write a PHP code to creating application for new bank account.

---------- Cut here ----------

8. (a) To develop a program in PHP for creating database, creating, Inserting, deleting table.

   (b) Write a PHP program to calculator.

DE–9567

Sub. Code
57

DISTANCE EDUCATION

M.Sc. (S.E.) (5 Years Integrated) DEGREE EXAMINATION, MAY 2017.

CASE TOOLS LAB

Time : Three hours Maximum : 100 marks

Examiner should select and give One question to each candidate by Lot system.

1. Develop an ATM System for a Bank with the following features :

   (a) The Customer has to login into the system using Credit Card number and Pin Number. The system should check for validation.

   (b) The System queries the customer for the type of Account either Savings Account or Current Account. After getting the type of Account, the system shows the amount left.

   (c) The System then queries the customer for the required amount. The user enters the amount and gets the money.

2. Online Quiz System has to be developed for conducting Preliminary stage of quiz as part of Department Technical symposium INTERRUPT The System developed should contain the following features

(a) The number of participants should be of 30.

(b) The duration for quiz is 30 minutes so a timer has to be kept which should show time duration.

---------- Cut here ----------

(c) The number of questions should be 40 chosen randomly from the database in three different areas namely

(i) Technical

(ii) Logical reasoning and

(iii) General Knowledge.

---------- Cut here ----------

(d) The questions should be of objective type with multiple options. For each correct answer the participant will receive 1 point and for wrong answer. 25 Point will be deducted.

---------- Cut here ----------

(e) At the end of the quiz the users' score along with information, whether he has been selected or not has to be displayed6. Once a student answered a question, he can't change the answer later.

3. Online ticket reservation system has to be developed for railway department. The System developed should contain following features

(a) The System should provide information about arrival and departure of trains along with information about stations through which it passes.

(b) Search about train passing through stations can be obtained either by means of train no, train name or specifying the source and destination stations.

(c) While displaying information about train, it has to provide following information's

(d) Stations through which train passes along with arrival and departure time.

(e) Availability of seats in different classes along with waiting list and RAC information.

(f) Based on the availability of tickets the ticket has to be issued. The ticked issued should contain the following information's PNR NO, Train No, Date, K.M., no of adults and children, Ticket No, Class, Ticket No, Coach, Seat/Berth, Sex, Age, T. Authority, Concession, Reservation fee, Total Cash, Train Name, Departure time.

4. Develop an Online stock maintenance system which should contain the following features. The stock items are of two types

   (a) Raw materials and

   (b) Finished products.

   There will be many vendors selling the raw materials and there will be many customers requiring the finished products. The procurement of raw materials is based on the number of products demanded. So there should be a balance between the raw materials and the finished products. Depending upon the need, every industry keeps a stock maintenance system. This system includes minimum stock, maximum stock based on the demand and supply. The proper record should be maintained at all entry and exit points.

---------- Cut here ----------

5. The telephone system provides simulation and automation of the telephone directory.

   The search module searches the given phone number in the database and retrieves the corresponding address. It concurrently displays multiple records whose data match the search key and allows indexed based and author based looping. It allows updates of telephone numbers that enable user to further update the details in the database. Most of these features require reusability and hence object oriented approach is used as the mode of development.